

TIC 66732476

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 3.108737 d
Epoch (t0) 2553.3754 BTJD
Duration 0.0840 d
Depth 9.814 % (98,141 ppm)
SNR 369.7

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 8 (gate ≥ 3)
Robust depth ratio 1.06 (gate ≥ 0.50)
Robust depth sigma 247.6 (gate ≥ 7.0)
Odd/even z-score 2.60 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 46 light-curve dip in this sector: 234.7 sigma t0 refined by -0.0050 phase
Difference-image peak SNR 26.93 (gate ≥ 5.0) Compactness 0.84 (gate ≥ 0.50)
Dimming centroid offset 0.36 px (8 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: none within the cutout

Phase-folded light curve (shaded = BLS eclipse window)

